Penghui Du

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Research Interests

I am curious about how individual brain networks support cognitive functions like language, memory, and emotion; how disruptions in these networks lead to neurological and psychiatric disorders; with the ultimate goal of identifying biomarkers and neuromodulation targets to advance personalized brain healthcare.

Education

24/08 - 27/03 **École Polytechnique Fédérale de Lausanne (EPFL)**, *Master in Neuro-X, GPA: 5.40 / 6.*

20/08 - 24/07 **Southern University of Science and Technology (SUSTech)**.

- Bachelor in Intelligent Medical Engineering, GPA: 3.84 / 4, Rank 2 / 22.

23/02 - 23/06 **University of Zurich (UZH)**, *Visiting Student in Neuroinformatics*.

Research

Harvard University, *Visiting Student*, Supervised by Dr. Randy Buckner.

26/03 - 27/03 I conceived and developed iGENMap, a novel precision mapping method that uses functional eigenmodes to map individual brain networks. It can resolve reliable individual network topography from as short as 14 min of fMRI, rather than the hours of data required by conventional methods. This enables population-scale studies of individual functional networks and substantially reduces the imaging burden for clinical applications.

26/03 - 27/03 Precision mapping offers accurate targets for TMS in depression, but also requires extensive fMRI data. I developed a novel framework for optimizing TMS target site and coil orientation in absence of individual fMRI data. Electric-field modeling showed that the framework achieves decent individual network selectivity with structural MRI alone or even scalp measurements alone, offering practical, low-burden options for clinical TMS targeting.

MPI for Human Cognitive and Brain Sciences, *Intern*, Supervised by Dr. Sofie Valk.

25/06 - 25/08 I developed a deep learning framework to predict individual brain glucose metabolism from structural and functional MRI features, demonstrating that metabolic profiles can be reliably inferred from MRI-based representations and generalizes across subjects and datasets.

Martinos Center for Biomedical Imaging, *Visiting Student*, Supervised by Dr. Jingyuan Chen.

23/07 - 23/12 We investigated the human brain resting-state metabolic connectivity derived from functional PET-FDG, and identified a robust superior-inferior metabolic architecture primarily driven by low-frequency dynamics, which aligned with known functional and anatomical organization.

Southern University of Science and Technology, *Research Assistant*, Supervised by Dr. Quanying Liu.

23/05 - 23/08 We reviewed how large language models can benefit from cognitive sciences, with the goal of integrating cognitive science priors into tests for artificial general intelligence.

22/09 - 24/07 We explored the effects of distinct drugs on large-scale neural dynamics of zebrafish brain using calcium imaging data collected from light sheet fluorescence microscopy.

Publications

26 / 11 Wan, B., **Du, P.**, Ferrari, A., Saberi, A., John, A., Yan, J., Liu, X., Hansen, J. Y., Liu, Z.-Q., Hong, S.-J., Carruzzo, F., Kaliuzhna, M., Geffen, T., Katthagen, T., Misic, B., Thompson, P. M., Van De Ville, D., Margulies, D. S., Warrier, V., Bethlehem, R. A. I., Schlagenhauf, F., Kaiser, S., Bernhardt, B., Valk, S. L., & Kirschner, M. EigenBrain maps cortical molecular architecture from functional, geometric, and structural eigenmodes. *bioRxiv* (2026). https://doi.org/10.64898/2026.10.27.758642

26 / 08 **Du, P.**, Coursey, S., Xu, T., Jamadar, S., Nolin, S., Wan, B., Wey, H., Polimeni, J., Price, J., Liu, Q., & Chen, J. Human Cerebral Cortex Organization Characterized by Functional PET-FDG "Metabolic Connectivity". *Eur J Nucl Med Mol Imaging* (2026). https://doi.org/10.1007/s00259-026-08109-5

26 / 03 **Du, P.** & Chen, J., Data-Driven Methods in Simultaneous EEG-fPET-fMRI: Opportunities and Challenges, at *2025 59th Asilomar Conference on Signals, Systems, and Computers*. https://doi.org/10.1109/IEEECONF67917.2025.11443781

25 / 04 Che, W., **Du, P.**, Huang, R., Liu, Q., Qu, Y., & Ye, Z. TUN-GCA: A Novel Approach for Organ Segmentation in Nasopharyngeal Carcinoma CT Images. In *International Workshop on Human Brain and Artificial Intelligence* (pp. 368–381). Springer Nature Singapore, 2024. https://doi.org/10.1007/978-981-96-4001-0_26

24 / 04 Qu, Y., Wei, C., **Du, P.**, Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., Wu, H., Liu, J., & Liu, Q. Integration of cognitive tasks into artificial general intelligence test for large models. *iScience*, 2024, 27(4), 109550. https://doi.org/10.1016/j.isci.2024.109550

24 / 03 Qu, Y., **Du, P.**, Wei, C., Che, W., Zhang, C., Ouyang, W., Bian, Y., Xu, F., Hu, B., Du, K., Wu, H., Liu, J., & Liu, Q. Promoting interactions between cognitive science and large language models. *The Innovation*, 2024, 5(2). https://doi.org/10.1016/j.xinn.2024.100579

21 / 11 Qu, Y., Jian, X., Che, W., **Du, P.**, Fu, K., & Liu, Q. Transfer learning to decode brain states reflecting the relationship between cognitive tasks. In *International Workshop on Human Brain and Artificial Intelligence* (pp. 110–122). Springer Nature Singapore, 2022. https://doi.org/10.1007/978-981-19-8222-4_10

Posters

26 / 11 **Du, P.**, Du, J., Billot, A., Sun, W., Yeo, B. T. T., Eldaief, M. C., & Buckner, R. L. iGENMap: Individualized Generative Mapping of Cortical Networks Using Functional Eigenmodes. **Late-Breaking Abstract**, *SfN 2026 Annual Meeting*, Washington, DC.

26 / 06 Wan, B., **Du, P.**, Ferrari, A., Saberi, A., John, A., Liu, X., Hansen, J., Liu, Z., Misic, B., Kaiser, S., Thompson, P., Warrier, V., Bethlehem, R., Bernhardt, B., Valk, S. & Kirschner, M., EigenBrain: a spatial framework for reconstructing individual brain molecular architecture from multimodal MRI, **Oral Presentation & Merit Abstract Award (6 out of 3240)** at *OHBM 2026 Annual Meeting*.

24 / 06 **Du, P.**, Coursey, S., Xu, T., Polimeni, J., Wey, H., Liu, Q., & Chen, J. Human Cerebral Cortex Organization Estimated by Functional PET-FDG Metabolic Connectivity, **Poster Presentation** at *OHBM 2024 Annual Meeting*.

Awards

25 / 02 EPFL/HMS **Bertarelli Fellowships (Five Recipients Per Year)**

24 / 07 SUSTech - **Distinguished Graduate Award**

22 / 10 SUSTech Department of Biomedical Engineering - **Outstanding Student Scholarships**

22 / 06 China College Student Innovation Training Program – **Undergraduate Research Grant**

22 / 06 2022 Guangdong Biomedical Engineering Design Competition – **First Prize**

22 / 05 13th "Challenge Cup" Guangdong Entrepreneurship Plan Competition – **First Prize**

19 / 05 The 13th Asia and Pacific Informatics Olympiad (APIO) – **Bronze Medal**

Talks

26 / 11 **Functional eigenmodes enable reliable mapping of individual-specific brain networks from limited data**, *Invited Lab Meeting Presentation, Precision Brain Mapping Lab @ ND*.

26 / 10 **Functional eigenmodes enable reliable mapping of individual-specific brain networks from limited data**, *Invited Lab Meeting Presentation, Speech Neurophysiology Lab @ UMich*.

26 / 09 **Functional eigenmodes enable reliable mapping of individual-specific brain networks from limited data**, *Invited Lab Meeting Presentation, Neuroscience and Experience of Social Transitions Lab @ UVA*.

26 / 03 **Human Cerebral Cortex Organization Characterized by Functional PET-FDG "Metabolic Connectivity"**, *at Molecular Connectivity Working Group (MCOS) Talk Series*.

24 / 02 **The Organization of Human Cerebral Cortex Estimated by Functional PET-FDG: The Promise and Controversy of Metabolic Connectivity** , *at Science on Tap*.

Teaching experience

22/02 - 23/02 **Introduction to Programming**, *Teaching Assistant and Workshop Organizer*, SUSTech.

Programming and Software Skills

Proficient: Connectome Workbench, Freesurfer, SimNIBS, SPM, FSL, Python, MATLAB, JAVA, LaTeX.

Working Knowledge: C/C++, AFNI, R, HTML, CSS, Git, Linux, Neurodesk.